

AN INFINITE FAMILY OF STRONGLY REGULAR DECOMPOSITIONS

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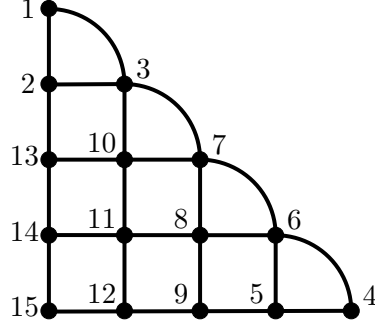
ABSTRACT. A strongly regular decomposition of a strongly regular graph is a partition of the vertex set into two parts on which the induced subgraphs are strongly regular, or cliques or cocliques. Strongly regular designs (SRDs) as defined by D.G. Higman in [13] are coherent configurations of rank 10 with two fibers in which the homogeneous configuration on each fiber is a strongly regular graph. Haemers and Higman in [11] proved the equivalence between strongly regular decompositions, excluding special cases, and strongly regular designs with certain parameter conditions. Here we define an infinite family of feasible parameters for pairs of strongly regular designs with fusion to strongly regular decomposition. These arise in graphs that are descendants of regular two-graphs, linking seven strongly regular graphs and three regular two-graphs within a single structure. This family includes the symplectic examples of [19], when the number of vertices is a power of two, and these SRDs are known to exist via a symplectic geometry construction. We investigate members of this family of orders 63, 255, 399 and prove non-existence of a strongly regular decomposition within these candidate graphs.

1. INTRODUCTION

The triangular graph $T(6)$ has vertex set consisting of the 15 2-sets from $\{1, 2, \dots, 6\}$ with adjacency given by non-empty intersection. From the graph depicted in Figure 1, it is easily seen that the vertices may be partitioned into two sets of sizes 6 (vertices 1 through 6) and 9 with induced graphs isomorphic to $2K_3$ and the lattice graph $L_2(3)$ respectively. All three of these graphs are strongly regular, allowing that $2K_3$ is imprimitive, and form what is called a *strongly regular decomposition*.

Shifting the focus from a single graph with two induced subgraphs to a more general configuration, this is an example of a *strongly regular design*. In that context, we view the partitioned vertex set as a set of vertices (say, the 6 vertices of the K_3 's) and a set of blocks (the other 9 vertices). We then define relations corresponding to (i) adjacent or (ii) non-adjacent points; (iii) adjacent or (iv) non-adjacent blocks; point-block incidence (v) or non-incidence (vi) given by adjacency in $T(6)$; and block-point incidence (vii) or non-incidence (viii) emphasizing that the last four are non-symmetric relations. The point graph and the block graph are both strongly regular. Adding an identity relation (ix) on points and another (x) on blocks, we have in total ten relations and they form a *coherent configuration*. Equivalently, the ten 0/1 adjacency matrices of these relations span a *coherent algebra* which by definition is closed under both Schur (entrywise) and matrix multiplication and contains the identity and the all-ones matrices.

Key words and phrases. strongly regular design, strongly regular decomposition, coherent configuration, strongly regular graph, Steiner system.

FIGURE 1. $T(6)$ 

The question of which strongly regular designs have mergings or *fusions* in which the union of relations i, iii, v, and vii is a strongly regular graph is the subject of [17]. This work considers a particular family of examples from that list that are related to regular two-graphs. To keep it fairly self-contained we include in Section 2 some background information on strongly regular graphs, regular two-graphs and strongly regular designs. For more on these structures we recommend [4], [9], [22] and [13].

In Section 3 we identify the family of interest. Each member of the family involves seven strongly regular graphs (not counting complements) and three regular two-graphs, all linked through a single graph containing two distinct strongly regular decompositions. Referring to the two strongly regular decompositions as SRD1 and SRD2, one main result is that SRD2 exists if and only if SRD1 exists. Our second main result is that these SRDs necessarily arise as subconstituents – that is, induced on the neighbours and non-neighbours of a vertex, respectively – of a larger strongly regular graph that represents a regular two-graph.

Two methods of construction are discussed in Section 4, and these are applied in Section 5 to three members of our family. We find no new strongly regular decompositions via these methods. Rather, we are able to enumerate those that do exist and to prove non-existence in a number of cases.

2. PRELIMINARIES

2.1. Strongly regular graphs. We recall some essential facts about strongly regular graphs and their parameters that will be used in this paper. For more background on this topic, the reader is referred to the standard references [4], [2], and [5].

Defn 2.1. A simple, undirected graph is strongly regular provided it is regular and the number of common neighbours to any two vertices depends only on whether they are adjacent.

The parameters of a strongly regular graph (SRG) will be denoted (v, k, λ, μ) where v is the number of vertices, k is the valency, and λ (μ) is the number of neighbours common to two adjacent (non-adjacent) vertices. We will use X for the vertex set of an SRG Γ , and A for the 0/1 adjacency matrix with rows and columns indexed by X . Let $\bar{A} := J - A - I$ where J is the all-ones matrix and I is the $v \times v$ identity matrix. This is the adjacency matrix of the complement $\bar{\Gamma}$ which is also strongly regular with parameters $(v, v - k - 1, v - 2k + \mu - 2, v - 2k + \lambda)$. The following are well known facts about these graphs.

Lemma 2.1. $A^2 = kI + \lambda A + \mu \bar{A}$.

Corollary 2.1. $\mathcal{A} := \text{span}(I, A, \bar{A})$ is closed under multiplication and is therefore an algebra, known as the Bose-Mesner or adjacency algebra of Γ .

The span is taken over $\mathbb{C}$ as is customary, although the results herein are valid over the rationals.

Defn 2.2. An SRG Γ is imprimitive if Γ or $\bar{\Gamma}$ is disconnected.

Supposing that Γ is disconnected, $\mu = 0$ so for any pair $x \sim y$, all other vertices are adjacent to both or neither x or y . This forces $\lambda = k - 1$, which implies the connected components of Γ are complete graphs. Lemma 2.1 then implies that the eigenvalues of A are k and -1 .

Lemma 2.2. If Γ is a primitive SRG then A has exactly three eigenvalues: k (with multiplicity 1), r , and s satisfying $r + s = \lambda - \mu$ and $rs = -(k - \mu)$. The multiplicities of r and s are:

$$f := \frac{k(s+1)(k-s)}{\mu(s-r)} \quad \text{and} \quad g := \frac{k(r+1)(k-r)}{\mu(r-s)}.$$

These multiplicities must of course be positive integers and this restricts the feasible parameter sets for SRGs.

Lemma 2.3. The algebra $\mathcal{A}$ is semi-simple with dual bases $\{I, A, \bar{A}\}$ and the set of minimal orthogonal idempotents, $\{E_0 = J/v, E_1, E_2\}$. Writing $A_0 = I$, $A_1 = A$, $A_2 = \bar{A}$, and $\bar{k} := v - k - 1$ for the valency of $\bar{\Gamma}$, the first and second eigenmatrices are P and Q/v where

$$P := \begin{bmatrix} 1 & k & \bar{k} \\ 1 & r & -r-1 \\ 1 & s & -1-s \end{bmatrix} \quad \text{and} \quad Q := \begin{bmatrix} 1 & f & g \\ 1 & fr/k & gs/k \\ 1 & -f(r+1)/\bar{k} & -g(s+1)/\bar{k} \end{bmatrix}.$$

Here $A_j = \sum_{i=0}^2 P_{ij} E_i$ and $E_j = \frac{1}{v} \sum_{i=0}^2 Q_{ij} A_i$.

Because $\mathcal{A}$ is closed under entrywise multiplication as well as matrix multiplication, we obtain another set of parameters from

$$E_i \circ E_j = \frac{1}{v} \sum_{k=0}^2 q_{ij}^k E_k.$$

Lemma 2.4 (The Krein conditions). *With the notation above, $q_{11}^1 \geq 0$ and $q_{22}^2 \geq 0$.*

These conditions lead to further restrictions on feasible parameters for SRGs.

By counting triples (x, y, z) of vertices with $x \sim y$, $x \sim z$, $y \not\sim z$ we obtain the identity:

Lemma 2.5. $k(k - \lambda - 1) = \mu(v - k - 1)$

2.2. Regular two-graphs.

Defn 2.3. *A two-graph is a set Ω of vertices along with a distinguished set Δ of triples from Ω such that every set of 4 vertices contains an even number of distinguished triples.*

We may associate a graph with a two-graph by defining Δ to be the set of triples carrying an odd number of edges of the graph. More formally, we may view a graph with vertex set X as a 2-coboundary $\Gamma : X^2 \rightarrow \mathbb{F}_2$ such that edges are precisely the pairs in the pre-image of 1. In that event, the standard coboundary operator $\delta : X^n \rightarrow X^{n+1}$ by

$$\delta f(B) := \sum_{x \in B} f(B \setminus \{x\})$$

maps the graph to a function on triples: in fact, $\delta\Gamma$ is a two-graph as can readily be checked, with the distinguished or *coherent* triples being the pre-image of 1. The set of coherent triples is fixed under the operation of Seidel switching, in which adjacencies and non-adjacencies of a select vertex x are interchanged: x is adjacent to 0, 1, or 2 vertices of any pair $\{y, z\}$. Upon switching, x is adjacent to 2, 1, or 0 of these respectively, hence the number of edges carried by the triple $\{x, y, z\}$ maintains its parity through switching. Sequentially switching on the vertices of any subset $U \subseteq X$ has the same effect as switching on the entire subset at once: adjacencies are affected only if they occur between U and $X \setminus U$.

A two-graph Ω may thus be defined as a switching class of graphs with vertex set X . Any graph Γ in this class is a representative of Ω , and we obtain the coherent triples as those carrying an odd number of edges of Γ . We write $[\Gamma]$ to denote the switching class of Γ .

While the 0/1 adjacency matrix is most frequently used to represent a simple graph algebraically, it is customary to use the Seidel matrix in the context of two-graphs. The Seidel matrix has rows and columns indexed by the vertices of the graph, with entries 0, -1 , or 1 according as the corresponding vertices are identical, adjacent, or not adjacent. Thus the Seidel and adjacency matrices are related by

$$S = J - 2A - I.$$

The reason that S is preferable in this context now becomes clear, as we see that switching on a subset U of the vertex set is equivalent to multiplying the corresponding rows and columns of S by -1 . This is accomplished by multiplying on the left and right by the diagonal matrix D whose i^{th} entry is -1 if $x_i \in U$ and 1 otherwise. Hence the new graph has Seidel matrix

$$S' = DSD$$

and it is immediately clear that switching preserves the spectrum of the Seidel matrix. We may therefore refer to the spectrum of a two-graph as determined by any of its representative graphs. A two-graph is *regular* if every pair of vertices is contained in the same number of

coherent triples. The parameter a represents this constant number of triples. We abbreviate regular two-graph as RTG.

An algebraic characterization of the regular two-graphs is found in Taylor ([22]) (cf Theorem 2.2). He proves that any representative of a regular two-graph is a *strong graph* with Seidel matrix having exactly two eigenvalues. A graph is *strong* if the number of vertices adjacent to exactly one of x and y depends only on whether $x \sim y$. Evidently, graphs that are both strong and regular are precisely the strongly regular graphs.

The following are well known results about regular two-graphs and their derived graphs ([22], [20]).

Given any representative graph of a two-graph Ω , we may isolate a vertex (by switching on its neighbours), to obtain another representative. If the isolated vertex is removed, we obtain a *descendant* graph of Ω .

Lemma 2.6. *Let Ω be a regular two-graph with v vertices and parameter a . Then Ω has a strongly regular descendant with SRG parameters*

$$(v - 1, a, (3a - v)/2, a/2) = (v - 1, 2\mu, 3\mu - v/2, \mu).$$

Conversely:

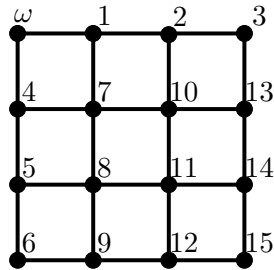
Lemma 2.7. *A strongly regular graph $\text{srg}(v, k, \lambda, \mu)$ is the descendant of a regular two-graph if and only if $k = 2\mu$.*

Thus, any SRG with $k = 2\mu$ defines a RTG $[\{\omega\} \cup \Gamma]$ with the addition of the isolated vertex ω . A related question is whether a RTG may have a representative that is strongly regular.

Lemma 2.8. *A strongly regular graph (v, k, λ, μ) determines a regular two-graph Ω as its switching class if and only if $v = 2(2k - \lambda - \mu)$. In this case, $a = 2(k - \mu)$ and the descendant of Ω is SRG $(v - 1, 2(k - \mu), k + \lambda - 2\mu, k - \mu)$.*

Example. Returning to our example of $T(6)$, observe that Lemma 2.7 applies to the SRG parameters $(15, 8, 4, 4)$ and thus $T(6)$ is a descendant graph.

FIGURE 2. $L_2(4)$



In fact, the RTG it descends from is the switching class of $L_2(4)$ which has SRG parameters $(16, 6, 2, 2)$. In Figure 2, isolate ω by switching on $N(\omega) = \{v_1, v_2, v_3, v_4, v_5, v_6\}$, to obtain $T(6)$ with the labelling shown in Figure 1. Observe that $N(\omega)$ as an induced subgraph is $2K_3$, which is Γ_1 in our notation as observed earlier.

For a RTG containing a strongly regular graph, the eigenvalues of the Seidel and adjacency matrices are related as shown in the two tables below. Here S has exactly two eigenvalues by assumption, so in the second table, $v - 2k - 1$ is equal to either $-2r - 1$ or $-2s - 1$ where r is the positive eigenvalue, and both possibilities occur.

$$(1) \quad \begin{array}{c|c|c|c} I & J & S & A = (J - S - I)/2 \\ \hline 1 & v & \rho_1 & (v - \rho_1 - 1)/2 \\ 1 & 0 & \rho_1 & (-\rho_1 - 1)/2 \\ 1 & 0 & \rho_2 & (-\rho_2 - 1)/2 \end{array} \quad \begin{array}{c|c|c|c} I & J & A & S = J - 2A - I \\ \hline 1 & v & k & v - 2k - 1 \\ 1 & 0 & r & -2r - 1 \\ 1 & 0 & s & -2s - 1 \end{array}$$

The next result shows that there are at most two SRGs in the switching class of a regular two-graph.

Lemma 2.9. *Let Γ be a strongly regular graph (v, k, λ, μ) such that $\Omega := [\Gamma]$ is a regular two-graph. Then exactly one other set of SRG parameters occurs in Ω .*

Proof. Suppose that Γ' is $\text{SRG}(v, k', \lambda', \mu')$ and is switching-equivalent to Γ . Let A' and S' be the adjacency and Seidel matrices of Γ' , respectively. Since S' has the same spec as S we see that the first table of (1) is unchanged, when S is replaced by S' and A by A' , with the possible exception that ρ_1 and ρ_2 are interchanged. In that case, the valency of Γ' is $k' := \frac{1}{2}v + s$, and the remaining two eigenvalues are $\{r', s'\} = \{r, s\}$.

By Lemma 2.2, $\mu' = k' + rs$ thus $\mu' = \mu + (k' - k)$. Similarly, $\lambda' = \mu' + r + s = \lambda + (k' - k)$. As $k' - k = (v/2 + s) - (v/2 + r)$, we find $\Gamma' = (v, \frac{1}{2}v + s, \lambda + s - r, \mu + s - r)$. $\square$

2.3. Strongly regular designs. For definitions and a table of parameters we refer to [13] and [17] although the essentials will be repeated here. For more on coherent configurations (CCs) and recent progress on small CCs, see [12] and [16].

Defn 2.4. *A strongly regular design is a finite incidence structure consisting of a set X_1 of points, a set X_2 of blocks, and an incidence relation $F \subseteq X_1 \times X_2$, such that the following are nonnegative integer constants:*

- $S_1 :=$ number of points incident with (in) each block;
- $S_2 :=$ number of blocks incident with (containing) each point;
- $a_1, b_1 :=$ the two distinct block intersection sizes;
- $a_2, b_2 :=$ the two distinct point join sizes, that is the number of blocks containing two given points;
- $N_1(P_1) :=$ number of points adjacent to a point x and incident with a block y , provided x is (is not) incident with y ;
- $N_2(P_2) :=$ number of blocks containing a point x and adjacent to a block y , given x is (is not) incident with y .

The two intersection sizes provide a natural way to define a graph Γ_i with vertex set X_i . Define two blocks to be adjacent or not according as they intersect in a_1 or b_1 points. Similarly, define the point graph Γ_1 with vertices X_1 such that a pair of adjacent (nonadjacent) points lies in a_2 (b_2) common blocks. It follows immediately from the definition that both Γ_1 and Γ_2 are strongly regular.

Let C be the 0/1 incidence matrix with rows indexed by the $v_1 := |X_1|$ points and columns by the $v_2 := |X_2|$ blocks. Let A_i be the 0/1 adjacency matrix of Γ_i . Then, letting context define the dimensions of J , we have ([13]):

- (i) C has row sum S_2 and column sum S_1 ;
- (ii) $CC^T = (S_2 - b_2)I + (a_2 - b_2)A_1 + b_2J$;
- (iii) $C^TC = (S_1 - b_1)I + (a_1 - b_1)A_2 + b_1J$;
- (iv) $CA_2 = (N_2 - P_2)C + P_2J$;
- (v) $A_1C = (N_1 - P_1)C + P_1J$.

Eigenvalues of Γ_i are determined as follows, with the multiplicities and remaining SRG parameters computed according to Lemma 2.2:

$$k_1 = \frac{S_2(S_1 - 1) - b_2(v_1 - 1)}{a_2 - b_2}, \quad \{r_1, s_1\} = \left\{ N_1 - P_1, -\frac{S_2 - b_2}{a_2 - b_2} \right\} \quad \text{and}$$

$$k_2 = \frac{S_1(S_2 - 1) - b_1(v_2 - 1)}{a_1 - b_1}, \quad \{r_2, s_2\} = \left\{ N_2 - P_2, -\frac{S_1 - b_1}{a_1 - b_1} \right\}.$$

Recall that an SRG is imprimitive if the graph or its complement is disconnected. Imprimitivity of a strongly regular design occurs when either there are repeated blocks, so that $S_1 = b_1$ or there are “repeated points” meaning $S_2 = b_2$. In the first case, we get $s_2 = -1$, and in the second $s_1 = -1$, which correspond to imprimitivity of the block graph and the point graph respectively.

We now have the pieces in place to confirm that an SRD is a *coherent configuration*, as defined below.

Defn 2.5. Let $\{A_i\}_{0 \leq i < r}$ be a set of 0/1-matrices with rows and columns indexed by a finite set X . Let $\mathcal{I} := \{0, 1, \dots, r-1\}$. The linear span $\mathcal{A} := \langle A_i \rangle_{\mathbb{C}}$ is a coherent algebra if:

- (i) $\sum_{i \in \mathcal{I}} A_i = J$,
- (ii) $\sum_{i \in \mathcal{L}} A_i = I$, for some subset $\mathcal{L} \subset \mathcal{I}$,
- (iii) for each i there exists $i^* \in \mathcal{I}$ such that $A_i^T = A_{i^*}$,
- (iv) $A_i A_j = \sum p_{ij}^k A_k$, $p_{ij}^k \in \mathbb{Z}^+$.

Each matrix A_i is equivalent to a relation on $X \times X$; the set X along with these relations is called a *coherent configuration* and abbreviated CC.

The set $\mathcal{L}$ induces a partition of the vertex set into *fibers*: for $i \in \mathcal{L}$, A_i is a 0/1 diagonal matrix. The 1's indicate the elements of one fiber. A consequence of the definition is that the A_i 's may be blocked according to these fibers such that no A_i is nonzero in more than two blocks, and not more than one if it is not symmetric. The *type* of a CC is a matrix indicating how many indices i appear in each block. A strongly regular design is a CC of type $\begin{bmatrix} 3 & 2 \\ 3 & 3 \end{bmatrix}$,

for example, which indicates two fibers X_1 and X_2 , with 3 relations on each representing the strongly regular point and block graphs, and 2 relations – incidence and non-incidence – on point-block pairs plus their transposes on block-point pairs. This CC therefore has *rank* $r = 10$.

2.3.1. Strongly regular decomposition. Let $\mathcal{S}$ be an SRD with the notation defined above. If there exists a fusion (that is, a merging of some of the 10 relations) to a strongly regular graph Γ with Γ_1 and Γ_2 as induced subgraphs, then Γ is a *strongly regular decomposition* as defined by Haemers and Higman ([11]). It is called *proper* if neither Γ_1 nor Γ_2 is a clique or coclique, and *exceptional* if the eigenvalues of Γ are distinct from those of Γ_1 and Γ_2 . It is the non-exceptional SRDs that are of interest in the work that follows.

Lemma 3.1 in [17], copied below, states parameter conditions under which a strongly regular decomposition occurs.

Lemma 2.10. ([17]) *Let $\mathcal{S}$ be an SRD as above, and let Γ be the graph obtained through fusion such that two vertices are adjacent if and only if they are adjacent in either Γ_1 or Γ_2 or are incident (point-block or block-point) in the design. Then Γ is strongly regular with strongly regular decomposition (Γ_1, Γ_2) if and only if:*

- (i) $k_1 + S_2 = k_2 + S_1$;
- (ii) $\lambda_1 + a_2 = \lambda_2 + a_1 = N_1 + N_2$;
- (iii) $\mu_1 + b_2 = \mu_2 + b_1 = P_1 + P_2$.

3. MAIN RESULTS

3.1. A family of feasible strongly regular decompositions. A table of feasible parameters is found in [17] for strongly regular designs that satisfy the conditions of Lemma 2.10. By observation from that table, we find a family dependent on a single parameter. The known symplectic graphs provide instances within this family, but can only account for cases in which the number of vertices is $2^n - 1$. In each case, there are two feasible SRDs the parameters of which are shown in the table below. We shall show that the two are related, in that the existence of one SRD implies existence of the other.

	Γ^+/Γ^-	Γ	SRD1		SRD2	
			Γ_1	Γ_2	Σ_1	Σ_2
v	$4m^2$	$4m^2 - 1$	$2m^2 - m$	$2m^2 + m - 1$	$2m^2 - m - 1$	$2m^2 + m$
k	$2m^2 \pm m$	$2m^2$	$m^2 - m$	m^2	m^2	$m^2 + m$
λ	$m^2 \pm m$	m^2	$\frac{m^2 - m}{2}$	$\frac{m^2 - m}{2}$	$\frac{m^2 + m}{2}$	$\frac{m^2 + m}{2}$
μ	$m^2 \pm m$	m^2	$\frac{m^2 - 2m}{2}$	$\frac{m^2}{2}$	$\frac{m^2}{2}$	$\frac{m^2 + 2m}{2}$
r	m	m	$-\frac{m}{2}$	$\frac{m}{2}$	$-\frac{m}{2}$	$\frac{m}{2}$
s	$-m$	$-m$	m	$-m$	m	$-m$
f	$2m^2 - m - 1^\dagger$	$2m^2 - m - 1$	$\frac{4m^2 - 4}{3}$	$\frac{4m^2 - 4}{3}$	$\frac{4m^2 - 4}{3}$	$\frac{4m^2 - 4}{3}$
g	$2m^2 + m^\dagger$	$2m^2 + m - 1$	$\frac{2m^2 - 3m + 1}{3}$	$\frac{2m^2 + 3m - 2}{3}$	$\frac{2m^2 - 3m - 2}{3}$	$\frac{2m^2 + 3m + 1}{3}$
S			m^2	$m^2 + m$	$m^2 - m$	m^2
a			$\frac{m^2 + m}{2}$	$\frac{m^2 + m}{2}$	$\frac{m^2 - m}{2}$	$\frac{m^2 - m}{2}$
b			$\frac{m^2}{2}$	$\frac{m^2 + 2m}{2}$	$\frac{m^2 - 2m}{2}$	$\frac{m^2}{2}$
N			$\frac{m^2 - m}{2}$	$\frac{m^2 + m}{2}$	$\frac{m^2 - m}{2}$	$\frac{m^2 + m}{2}$
P			$\frac{m^2}{2}$	$\frac{m^2}{2}$	$\frac{m^2}{2}$	$\frac{m^2}{2}$

$^\dagger f$ and g are interchanged for Γ^- .

We refer to the graphs of SRD 1 as Γ , with decomposition into Γ_1 and Γ_2 . For the second SRD we use the notation Γ , Σ_1 , and Σ_2 . The following lemma is immediate from Lemmas 2.5 through 2.9.

Lemma 3.1. *Let $\mathcal{S}_1$ be an instance of SRD 1 and $\mathcal{S}_2$ an instance of SRD 2 as in the table above. Then:*

- (i) Γ is the derived graph of a regular two-graph Ω with $v = 4m^2$ and $a = 2m^2$;
- (ii) Ω contains two possible SRGs we will denote $\Gamma^\pm$, namely

$$(4m^2, 2m^2 \pm m, m^2 \pm m, m^2 \pm m);$$
- (iii) Γ_1 determines a RTG Ω_1 , with descendant Σ_1 ;
- (iv) Σ_2 determines a RTG Ω_2 , with descendant Γ_2 .

We conclude that each instance of this family links seven SRGs and three RTGs. All seven of the SRGs are either descendants or representatives of a RTG. We remark that aside from the restriction $m \equiv \pm 2 \pmod{6}$, all SRGs presented here are feasible in the sense that the Krein conditions are met and other SRG identities are satisfied. One exception is the case $m = 2$: there is an SRD with graphs K_5 and Σ_2 . However, Σ_1 in the table is not valid for $m = 2$.

3.2. Existence of the SRGs and RTGs. Individual instances of this family up to $m = 16$ are shown in Section 6 with information about existence of the SRGs involved. The Γ^+ graphs

may be derived from regular symmetric Hadamard matrices with constant diagonal (RSHCDs), where the Hadamard matrix is $I + S$ ([4], [10], [23]). In addition, Γ^- has latin square type $LS_m(2m)$ while Γ^+ has negative latin square type $NL_m(2m)$.

A construction for Γ^ϵ ($\epsilon = \pm$) is detailed in [7]. We summarize the method here as it is relevant to the examples presented in Section 5.

- (1) The construction uses a symmetric, $v \times v$, real Hadamard matrix H and a $b \times v$ incidence matrix B of a BIBD (v, k, λ, r, b) . We used Hadamard matrices found at [21] and the $(v, 2, 1, v-1, \binom{v}{2})$ BIBD with all $\binom{v}{2}$ 2-elements subsets of $\{1, 2, \dots, v\}$ as blocks. Rows of B are sorted so that the first $v/2$ rows sum to j^T , representing $v/2$ disjoint blocks.
- (2) One row of H is removed, forming the $(v-1) \times v$ matrix H' .
- (3) A $\binom{v}{2} \times v^2$ matrix Φ is constructed by sequentially replacing the $v-1$ nonzero entries of each column of B with the $v-1$ rows of H' ; and replacing each 0 of B with a row of v 0's.
- (4) The columns of Φ contain $v-1$ entries equal to ± 1 with remaining entries 0. Thus the diagonal entries of $\Phi^T \Phi$ are all equal to $v-1$. The non-diagonal entries are all ± 1 because there is exactly one row of B (and hence Φ) in which two given columns are both nonzero. Then

$$S := -(\Phi^T \Phi) + (v-1)I$$

is the Seidel matrix of a graph which is in fact an SRG with the parameters of Γ^ϵ . We obtained Γ^- when row 1 was removed from H and Γ^+ when any other row was removed.

Examples of the descendant graphs are the complements of the symplectic graphs for the cases in which m is a power of 2. In addition, these graphs relate to Steiner systems: $S(2, k, 2k^2 - k)$ for k even, which are equivalent to $2 - (2k^2 - k, k, 1)$ designs. Given an incidence structure consisting of a set of $2k^2 - k$ points and a set of blocks of size k such that any two points are in exactly one block, elementary constraints on 2-designs, namely that $bk = vr$ and $\lambda(v-1) = r(k-1)$, imply that b , the number of blocks, is $4k^2 - 1$ and that r , the number of blocks incident with each point, is $2k + 1$. Furthermore, two distinct blocks must either be disjoint or meet in one point. We claim both of these intersection sizes occur, because otherwise the design is symmetric which forces $k = r$ and that is impossible. Thus, these 2-designs are quasi-symmetric, having two block intersection sizes, and from this we obtain the fact that the block graph (for either intersection size) is strongly regular. The strongly regular parameters are those of Γ for blocks that are adjacent if and only if disjoint.

Note that the Steiner block graph is Γ here, whereas the block graph of the SRD is Γ_2 . These Steiner systems are known for k a power of 2 but unknown for $k \equiv 2 \pmod{4}$ ([4], 8.5.4 and [6]).

3.3. Main theorems. Our main results are presented in three theorems. The first is that SRD 1 and SRD 2 are related under Seidel switching and implies that existence or non-existence of one suffices to decide that of the other.

Our second result establishes conditions under which an SRG is obtained upon switching $\Gamma \cup \{\omega\}$ on the vertices of Γ_1 . When applied to our family, the graph obtained is necessarily one of $\Gamma^\pm$. This has the important consequence that Γ_1 and Γ_2 may be constructed as subconstituents of this graph – that is, Γ_1 is the induced graph on the neighbourhood of a point in $\Gamma^\pm$, and Γ_2 is induced on the non-neighbours. This leads directly to our first method of construction, detailed in Section 4.1 with an extended example. The final theorem asserts that SRDs in this family only exist when defined by strongly regular subconstituents of Γ^ϵ .

Theorem 3.2. *In the notation of the previous section, suppose Γ admits a strongly regular decomposition equivalent to SRD1. Then SRD2 exists and may be obtained by switching on the neighbourhood (in $\Gamma \cup \{\omega\}$) of a vertex of Γ_1 .*

Proof. Let Γ' be the graph obtained from Γ by adjoining an isolated vertex ω . From the parameters of Γ_1 we see that the condition $v = 2(2k - \lambda - \mu)$ is met, giving this graph a strongly regular descendent $\tilde{\Gamma}_1$ with order $2m^2 - m - 1$. By Lemma 2.8, this is Σ_1 of SRD 2. Let $x \in X_1$ and let N be the neighbourhood of x in Γ' . We isolate x by switching Γ' on N to obtain $\tilde{\Gamma}$ and this has two consequences: firstly, X_1 now carries the descendent graph $\tilde{\Gamma}_1$ plus $\{x\}$; secondly ω is now adjacent to all vertices of N in $\tilde{\Gamma}$.

Since $\tilde{\Gamma}$ lies in the switching class of Γ' and has an isolated vertex, the remaining $4m^2 - 1$ vertices form another SRG Γ . It remains to show that the graph on $X_2 \cup \{\omega\}$ is strongly regular with the parameters of Σ_2 .

To show the graph is regular, we treat three cases, considering first the vertex ω . After x is isolated, ω is adjacent to $|N(x) \cap X_2| = S_2$ vertices of X_2 .

Now consider $y \in X_2$ and $y \sim x$ in Γ' . Since y lies in the switching set, it has (after switching) $1 + N_2 + (v_2 - k_2 + N_2 - S_2)$ neighbours in X_2 as can be seen in Figure 3.

Next, consider $y \in X_2$ and not adjacent to x in Γ' as shown in Figure 4. In this case, y is not in the switching set so has $(k_2 - P_2) + (S_2 - P_2)$ neighbours in X_2 after switching.

Thus we require these valencies to be equal for $\tilde{\Gamma}_2$ to be regular. That is,

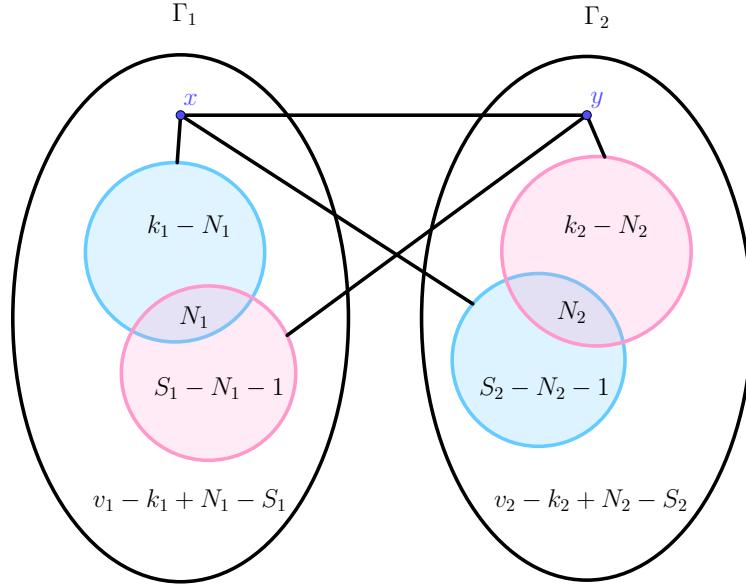
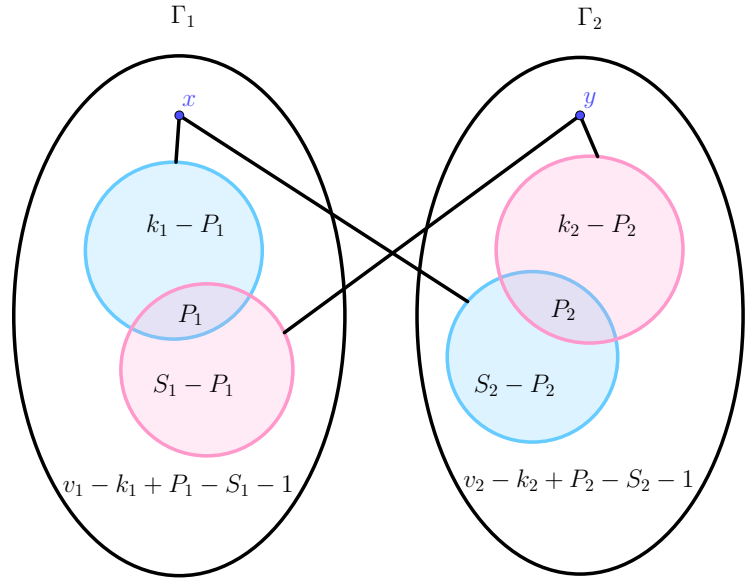
$$S_2 = v_2 - k_2 - S_2 + 2N_2 + 1 = k_2 + S_2 - 2P_2.$$

Since all of these quantities equal $m^2 + m$ for this family of parameters, $\tilde{\Gamma}_2$ is regular.

That $\tilde{\Gamma}_2$ is strongly regular follows from the observation that it is contained in $[\Gamma_2 \cup \{\omega\}]$ and this switching class is a RTG by Lemma 3.1. It follows that all graphs in the switching class are strong, hence all such regular graphs are strongly regular.

To confirm that the parameters match those of Σ_2 , apply Lemma 2.8. Any SRG (v, k, λ, μ) with descendant Γ_2 must satisfy:

$$\begin{aligned} v &= 2m^2 + m \\ k - \mu &= \frac{m^2}{2} \\ k - 2\mu + \lambda &= \frac{m^2 - m}{2}. \end{aligned}$$

FIGURE 3. $y \sim x$ in Γ' FIGURE 4. $y \not\sim x$ in Γ' 

We have seen that this graph is regular with valency $m^2 + m$, thus $\mu = \frac{m^2}{2} + m$. It follows that $\lambda = \frac{m^2+m}{2}$ and we have Σ_2 . $\square$

Example. To illustrate this with $L_2(4)$, recall that $T(6)$ is a descendant of this graph, and that we obtain an SRD of the descendant graph by switching on $N(\omega)$ in $L_2(4)$. The labelling of both graphs is as in Figures 1 and 2. To see the second SRD in this example, observe that we may now switch $T(6) \cup \{\omega\}$ on $N(v_1) = \{v_2, v_3, v_{10}, v_{11}, v_{12}, v_{13}, v_{14}, v_{15}\}$. The result is that ω is now adjacent to the 8 elements of that switching set, taking the place of v_1 . Secondly, the induced graph on $\{v_4, v_5, \dots, v_9\}$ is $T(4)$ both before and after switching, with the neighbourhoods of v_4 and v_7 interchanged, as well as those of v_5 and v_8 ; and v_6 and v_9 . The partition $\{\{v_2, v_3, v_4, v_5, v_6\}, \{\omega, v_7, v_8, \dots, v_{15}\}\}$ now gives an SRD inside another copy of $T(6)$ in which the first induced graph is K_5 and the second is $T(5)$, the complement of the Petersen graph.

Theorem 3.3. *Let $\mathcal{S}$ be an SRD with the notation of Section 3.1. In particular, let Γ be such that $[\Gamma \cup \{\omega\}]$ is a regular two-graph. Then switching $\Gamma \cup \{\omega\}$ on X_1 results in a strongly regular graph Γ^+ if and only if (i) $k_2 = S_1$ and (ii) $v_1 = v_2 + k_1 - S_2 + 1$. Furthermore, a similar result holds with the subscripts 1 and 2 interchanged.*

Proof. The adjacency matrix of $\Gamma \cup \{\omega\}$ has the form

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & A_1 & C \\ 0 & C^T & A_2 \end{bmatrix}$$

where the blocks are indexed by $\{\omega\}$, X_1 , and X_2 . After switching on X_1 , we have an adjacency matrix of the form

$$\begin{bmatrix} 0 & j^T & 0 \\ j & A_1 & J - C \\ 0 & J - C^T & A_2 \end{bmatrix}.$$

Recall that j is the all-ones vector of appropriate length and J is the all-ones matrix of appropriate dimensions. This graph is regular if and only if the matrix above has constant row sum. As the first row has sum v_1 , the next v_1 rows have sum $1 + k_1 + v_2 - S_2$, and the bottom v_2 rows have sum $v_1 - S_1 + k_2$, we see that the stated conditions are both necessary and sufficient for regularity of Γ^+ . Because Γ^+ is regular and lies in the switching class of a RTG, it is strongly regular. $\square$

Clearly the conditions of Theorem 3.3 are met by both SRD1 and SRD2.

We have seen that an SRG Γ with a strongly regular decomposition from the partition $X_1 \cup X_2$ can be obtained from an SRG Γ^+ in $[\Gamma \cup \{\omega\}]$ by switching on the neighbourhood of a vertex x , and that in this instance X_1 and X_2 are necessarily the neighbours and non-neighbours of x . In other words, Γ^+ has strongly regular sub-constituents. The next theorem addresses the question of whether these SRDs must arise in this way: that is, is it possible for a descendant graph Γ to contain an SRD with vertex sets X_1 and X_2 , such that X_i is not the neighbourhood of a point in Γ^+ ? There are graphs with these parameters that do not have strongly regular

subconstituents as we shall see in Section 5. Our aim is to prove that this does not happen in cases where an SRD is present in the derived graph Γ .

Proposition 3.1. *Let Ω_1 and Ω_2 be SRGs (v, k, λ, μ) . Then Ω_1 and Ω_2 are switching equivalent if and only if they are identical.*

Proof. Let $S \subseteq X$ and suppose Ω_2 is obtained from Ω_1 by switching on S . Let x be a vertex in $\bar{S} = X \setminus S$. We claim:

$$|N(x) \cap S| = \frac{1}{2}|S|.$$

To see this, let $a = |N(x) \cap S|$ so that x has $k - a$ neighbours in $\bar{S}$. After switching, x is adjacent to $|S| - a$ vertices in S and $k - a$ in $\bar{S}$. By regularity we obtain $|S| = 2a$ as desired.

Next, we show that for any pair $x, y \in \bar{S}$, $N(x) \cap S = N(y) \cap S$. That is, x and y are adjacent to the same half of S . First, consider $x \sim y$. Let b be the number of common neighbours to x and y in S . Then $\lambda - b$ vertices of $\bar{S}$ are adjacent to both x and y . After switching, there are $2a - b$ common neighbours in S and $\lambda - b$ in $\bar{S}$. As these must total to λ we obtain $a = b$ so that all neighbours of x in S are also neighbours of y .

Now consider $x \not\sim y$. Letting $c = |N(x) \cap N(y) \cap S|$, note that after switching x and y have $2a - c$ common neighbours in S and $\mu - c$ in $\bar{S}$. Equating the total to μ we find $c = a$.

At this point we have all vertices of $\bar{S}$ adjacent to a fixed half of S . Let z be a vertex in this half of S . Clearly z has $v - 2a$ neighbours in $\bar{S}$ before switching, and 0 after since this is all of $\bar{S}$. Yet there is no change to adjacencies within S . This contradicts regularity unless $v = 2a$, in which case the switching set is all of X and Ω_1 is identical to Ω_2 . $\square$

Theorem 3.4. *With the notation of Section 3.1, Γ has a strongly regular decomposition if and only if Γ^+ and Γ^- have strongly regular subconstituents.*

Proof. It is immediate that we obtain a strongly regular decomposition of Γ by switching on the neighbourhood of a vertex ω in Γ^ϵ , given that the first and second subconstituents of the latter graphs are SRGs.

We now suppose that Γ has an SRD, $\mathcal{S}$. By Theorems 3.2 and 3.3 we may assume that $\mathcal{S}$ is SRD1. We know from Theorem 3.3 that switching $\Gamma \cup \{\omega\}$ on X_1 (without loss of generality) yields an SRG Ω_1 with the parameters of Γ^+ . Clearly, that graph has strongly regular subconstituents. What must be demonstrated is that every other SRG with Γ as a descendant does as well. To that effect, suppose that switching $\Gamma \cup \{\omega\}$ on some set W yields a strongly regular graph Ω_2 with the parameters of Γ^+ . As Ω_2 is switching equivalent to Ω_1 via the switching set

$$U := (W \cup X_1) \setminus (W \cap X_1),$$

these graphs are identical by Proposition 3.1. $\square$

Remark: In light of Theorem 3.2 we wish to clarify that Γ^+ and Γ^- are switching equivalent and both have Γ as a descendant yet they are not identical. This does not contradict Proposition 3.1 because their parameters are distinct and in fact the descendant graphs are not the same. Yet, it is true that both Γ^+ and Γ^- must have strongly regular subconstituents if one of them

does. This is because switching Γ^+ on $X_1 := N(\omega)$ defines, say, SRD1 in $\{\omega\} \cup \Gamma$; switching as in Theorem 3.2 yields SRD2 in $\{\omega'\} \cup \Gamma'$ which in turn shows that Γ^- has strongly regular subconstituents, namely $\tilde{X}_1$ and $\tilde{X}_2$.

4. CONSTRUCTION

4.1. Method 1: Decomposition via switching. Theorems 3.2 and 3.3 provide a method for constructing the SRDs in our family, provided they exist and we have the SRG Γ^+ or any strongly regular representative of the RTG $[\Gamma^+]$. Isolate a vertex ω by switching on $N(\omega)$; remove ω to obtain Γ . The switching set forms the vertex set of Γ_1 , and this of course determines Γ_2 as well. This method may also be used to prove a given Γ does not have SRD1 or SRD2. When Γ^+ is not vertex-transitive, it may be necessary to check every descendant graph for possible decompositions.

4.2. Method 2: Decomposition via searching an eigenspace. This method is applicable to searching for SRDs in general, thus is independent of our family of candidates related to RTGs. It may also be used to enumerate the decompositions as distinct partitions of the vertices of Γ . We note that what follows includes a correction to the proof of Lemma 4.3 in [17].

Lemma 4.1. *Let $\mathcal{S}$ be a strongly regular design with fusion to strongly regular graph Γ . The adjacency matrix of Γ has an eigenvector with v_1 entries equal to S_2 and v_2 entries equal to $-S_1$. The corresponding eigenvalue is $k_1 + k_2 - k$.*

Proof. Because the all ones vector j_i , of length v_i , is an eigenvector for Γ_i ($i = 1, 2$), we consider an eigenvector for Γ of the form $w := \begin{bmatrix} j_1 \\ a j_2 \end{bmatrix}$, where vertices of Γ are ordered so that A has block form

$$\begin{bmatrix} A_1 & C \\ C^T & A_2 \end{bmatrix}.$$

Recall that the row and column sums of C are S_2 and S_1 respectively. We find:

$$\begin{aligned} A_1 j_1 + a C j_2 &= \alpha j_1 &\implies & k_1 + a S_2 = \alpha, \\ C^T j_1 + a A_2 j_2 &= \alpha a j_2 &\implies & \frac{1}{a}(S_1 + a k_2) = \alpha \end{aligned}$$

where α is a purported eigenvalue for w . Thus $S_1 + a k_2 = a(k_1 + a S_2)$ which is quadratic in a . This has $a = 1$ as one solution due to regularity of Γ ; the other solution is $a = -S_1/S_2$ which equates to $-(k - k_2)/(k - k_1)$. The eigenvalue of Γ is $k_1 - S_1 = k_1 + k_2 - k$. This is the conclusion of Lemma 4.3 in [17], but the proof given therein is only correct when $S_1 = S_2$ which happens when $k_1 = k_2$. $\square$

Remark. If working in the complement of Γ is necessary, we obtain $a = -(n_1 - S_1)/(n_2 - S_2)$ by the same argument.

We now outline a method of searching for a strongly regular decomposition of Γ by finding all eigenvectors meeting the description given in the lemma. This method is adapted from [3] (see 3.13). The v_1 indices with entries equal to S_2 represent the vertices of Γ_1 , and so each

such vector determines a partition of X into two sets on which the induced graphs are regular. (However, this does not guarantee they are strongly regular.)

From the primitive idempotent E_j we have a basis of the j^{th} eigenspace – namely, the column space of that matrix. Assuming we are searching for a non-exceptional SRD, the eigenvalues of Γ are integers and thus the appropriate eigenspace has a basis over the rationals. We may easily use row reduction to obtain a basis consisting of the rows of a matrix of the form $[I_n \ B]$ where $n \in \{f, g\}$. There are 2^n vectors in the row space for which the first n entries, at least, lie in $\{S_2, -S_1\}$. Testing the remaining entries of each of those, we obtain the full set of eigenvectors of the form $S_2^{v_1}, (-S_1)^{v_2}$.

The final step in this method is to test whether the partition determined by each of these eigenvectors supports a strongly regular decomposition. This was tested for our examples using the GAP packages AssociationSchemes and COCO2P ([8], [1], [14]).

5. EXAMPLES

GAP code and adjacency matrices for these examples can be found in the GitHub repository [18].

Case $m = 2$. We have seen that both SRDs may be constructed via switching from the RTG on 16 vertices with $L_2(4) = \text{SRG}(16, 6, 2, 2)$ as representative. An eigenspace search reveals 70 eigenvectors in $\{3, -2\}^{15}$ in the (-2) -eigenspace of Γ . Exactly 10 of these are confirmed by computer testing to give SRD1. This is consistent with a simple counting argument: select a pair of disjoint K_3 's in $T(6)$ by first choosing two adjacent vertices in $15 \cdot 8/2$ ways, then select the unique common neighbour that is not part of a 5-clique, to form a K_3 . There is then a unique K_3 disjoint from this one, and the remaining nine vertices will form Γ_2 . Since we can select the same pair of K_3 's in 6 ways, we see that there are 10 distinct SRDs of this type. All are isomorphic as the graphs on 15, 5, and 9 vertices are unique.

Case $m = 4$.

- (1) The halved, folded 8-cube is $\text{SRG}(64, 28, 12, 12)$. Isolating a vertex, we obtain SRD1 on $\text{SRG}(63, 32, 16, 16)$ and this graph is the complement of the symplectic graph. Thus we construct both SRD1 and SRD2 by Method 1. Enumeration by Method 2 revealed 841,032 vectors in the 3-eigenspace of the form $4^{35}(-5)^{28}$. Of those, 36 give valid SRDs.
- (2) A fusion of the 6-cube produces this same graph: namely, the Hamming scheme $H(6, 2)$ has a fusion according to the partition $[[0], [1, 4, 5], [2, 3, 6]]$, resulting in $\text{SRG}(64, 28, 12, 12)$ and again we find the descendant graph isomorphic to the symplectic graph $Sp(6, 2)$. This belongs to family of such examples detailed in [19].
- (3) Following the method of Fickus et al in [7] adapted from [10], and outlined in Section 3.2, we construct $\text{SRG}(64, 36, 20, 20)$ or $\text{SRG}(64, 28, 12, 12)$ depending on which row of the Hadamard matrix of order 8 is deleted. All 8 cases were tested using GAP. In each case, we checked each of the 64 vertices for strongly regular subconstituents. In no case did Method 1 reveal a valid SRD, leading to the conclusion by Theorem 3.4 that these graphs do not have strongly regular decompositions. Of the 8 SRGs on 64

points, we found automorphism groups of sizes 1, 2, and 4. Isomorphism testing in GAP indicated that no two of these 8 graphs are isomorphic. In testing each of the 64 vertices in each of the 8 SRGs we constructed 512 descendant graphs at least 8 of which are pairwise non-isomorphic as determined by distinct, small automorphism groups. It is likely that these fall among the many such graphs referenced in [4], “most of them ugly”.

- (4) Following Klin et al in [15], we constructed a non-Schurian SRG $(63, 30, 13, 15)$ as a fusion of a rank 6 coherent configuration afforded by the action of $P\Gamma L(2, 8)$ on cosets of a subgroup of order 24. Method 2 was employed on this graph, with the result that 840 eigenvectors of the form $\{4, -3\}^{63}$ were found in the (-5) eigenspace. However, none of these produced a strongly regular decomposition.

In summary, only in the symplectic graphs do we find strongly regular decompositions. We have shown non-existence in 9 other graphs with the same parameters.

Case $m = 8$.

- (1) Construction via a fusion of the 8-cube produces a RTG on 256 vertices. Namely, the Hamming scheme $H(8, 2)$ has a fusion according to the partition $[[0], [1, 4, 5, 8], [2, 3, 6, 7]]$ ([19]) that results in $\Gamma = (256, 120, 56, 56)$ which affords a RTG. The descendant graph is $\Gamma = (255, 128, 64, 64)$ and is isomorphic to the symplectic graph of $Sp(8, 2)$. The decomposition of this graph into $\Gamma_1 = (120, 56, 28, 24)$ and $\Gamma_2 = (135, 64, 28, 32)$ is obtained from the first and second subconstituents of Γ , where Γ_1 is the induced graph on the neighbourhood of vertex 1.
- (2) The halved, folded 10-cube is isomorphic to this fusion of the 8-cube. More generally, the halved, folded $2n$ -cube is isomorphic to a fusion of the $(2n - 2)$ -cube.
- (3) The Fickus et al method was employed using the unique symmetric Hadamard matrix of order 16, to produce 16 SRGs on 256 vertices. As in the $m = 4$ case, none of these graphs have strongly regular subconstituents as determined by computing the $16 \cdot 256$ descendants and therefore by Theorem 3.4 no SRDs exist in these graphs.

Case $m = 10$. The construction of [7] was again used to search for SRDs on 400 points. As there are two inequivalent, symmetric Hadamard matrices of order 20 ([21]), there were 40 SRGs to investigate with parameters $(400, 210, 110, 110)$ or $(400, 190, 90, 90)$. In each case, each of the 400 vertices was isolated to produce a descendant graph on 399 points. In no case did Method 1 reveal an SRD. It is not known at this time how many isomorphism classes are represented among the 40 graphs constructed on 400 vertices, or the 16,000 descendants.

6. TABLES

	v	k	λ	μ	r	s	f	g	comments
Γ^+	16	10	6	6	2	-2	5	10	! Clebsch
Γ^-	16	6	2	2	2	-2	6	9	2! Shrikhande, RSHCD ⁺
Γ	15	8	4	4	2	-2	5	9	! $T(6)$
Γ_1	6	2	1	0	-1	2	4	1	!, imprimitive
Γ_2	9	4	1	2	1	-2	4	4	! $L_2(3)$
Σ_1	5	4	3	-	-	-1	-	4	! K_5
Σ_1	10	6	3	4	1	-2	4	5	! $T(5)$

	v	k	λ	μ	r	s	f	g	comments
Γ^+	64	36	20	20	4	-4	27	36	+ (Fickus et al)
Γ^-	64	28	12	12	4	-4	28	35	+ RSHCD ⁺
Γ	63	32	16	16	4	-4	27	35	+ (many)
Γ_1	28	12	6	4	-2	4	20	7	4! $T(8)$, Chang
Γ_2	35	16	6	8	2	-4	20	14	3853! (McKay & Spence)
Σ_1	27	16	10	8	4	-2	6	20	! Schläfli
Σ_2	36	20	10	12	2	-4	20	15	32548! comp (McKay & Spence)

	v	k	λ	μ	r	s	f	g	comments
Γ^+	256	136	72	72	8	-8	119	136	+ (Fickus et al)
Γ^-	256	120	56	56	8	-8	120	135	+ RSHCD ⁺
Γ	255	128	64	64	8	-8	119	135	+
Γ_1	120	56	28	24	-4	8	84	35	+
Γ_2	135	64	28	32	4	-8	84	50	+ comp of $O^+(8, 2)$
Σ_1	119	64	36	32	8	-4	34	84	+ comp of $O^-(8, 2)$
Σ_2	136	72	36	40	4	-8	84	51	+

	v	k	λ	μ	r	s	f	g	comments
Γ^+	400	210	110	110	10	-10	189	210	+ (Fickus et al)
Γ^-	400	190	90	90	10	-10	190	209	+ RSHCD ⁺
Γ	399	200	100	100	10	-10	189	209	+
Γ_1	190	90	45	40	-5	10	132	57	?
Γ_2	209	100	45	50	5	-10	132	76	+
Σ_1	189	100	55	50	10	-5	56	132	?
Σ_2	210	110	55	60	5	-10	132	77	+ (comp of Klin et al)

	v	k	λ	μ	r	s	f	g	comments
Γ^+	784	406	210	210	14	-14	377	406	+ (Fickus et al)
Γ^-	784	378	182	182	14	-14	378	405	+ RSHCD ⁺
Γ	783	392	196	196	14	-14	377	405	+
Γ_1	378	182	91	84	-7	14	260	117	?
Γ_2	405	196	91	98	7	-14	260	144	?
Σ_1	377	196	105	98	14	-7	116	260	?
Σ_2	406	210	105	112	7	-14	260	145	?

	v	k	λ	μ	r	s	f	g	comments
Γ^+	1024	528	272	272	16	-16	495	528	+ (Fickus et al)
Γ^-	1024	496	240	240	16	-16	496	527	+ RSHCD ⁺
Γ	1023	512	256	256	16	-16	595	527	+
Γ_1	496	240	120	112	-8	16	340	155	+
Γ_2	527	256	120	128	8	-16	340	186	+ comp of $O^+(10, 2)$
Σ_1	495	256	136	128	16	-8	154	340	+ comp of $O^-(10, 2)$
Σ_2	528	272	136	144	8	-16	340	187	+

REFERENCES

1. J. Bamberg, A. Hanaki, and J. Lansdown, *AssociationSchemes, a gap package for working with association schemes and homogeneous coherent configurations*, Vversion 2.1.0, <http://www.jessielansdown.com/AssociationSchemes>, Jun 2022, GAP package.
2. A. E. Brouwer, *Parameters of strongly regular graphs*, <https://www.win.tue.nl/~aeb/graphs/srg/srgtab.html>, Accessed: Aug 2026.
3. A. E. Brouwer and W. H. Haemers, *Spectra of Graphs*, Springer, New York, 2012.
4. A. E. Brouwer and H. Van Maldeghem, *Strongly Regular Graphs*, Encyclopedia of Mathematics and its Applications, vol. 182, Cambridge University Press, Cambridge, 2022.
5. P. J. Cameron and J. H. van Lint, *Designs, Graphs, Codes and their Links*, London Math. Soc. Student Texts, vol. 22, Cambridge University Press, Cambridge and New York, 1991.
6. R. H. F. Denniston, *Some maximal arcs in finite projective planes*, J. Combinatorial Theory **6** (1969), 317–319.
7. M. Fickus, J. Jasper, D. G. Mixon, and J. D. Petersen, *Equiangular tight frames with centroidal symmetry*, Appl. Comput. Harmon. Anal. **44** (2018), 476–496.
8. *GAP-Groups, Algorithms, and Programming*, Version 4.12.2, <https://www.gap-system.org>, Dec 2018.
9. C. D. Godsil, *Algebraic Combinatorics*, Chapman and Hall Mathematics Series, Chapman & Hall, New York, 1993.
10. J.-M. Goethals and J. J. Seidel, *Strongly regular graphs derived from combinatorial designs*, Canadian J. Math. **22** (1970), 597–614.
11. W. H. Haemers and D. G. Higman, *Strongly regular graphs with strongly regular decomposition*, Linear Algebra Appl. **114/115** (1989), 379–398.
12. D. G. Higman, *Coherent algebras*, Linear Algebra Appl. **93** (1987), 209–239.
13. ———, *Strongly regular designs and coherent configurations of type $\begin{bmatrix} 3 & 2 \\ 3 & 3 \end{bmatrix}$* , European J. Combin. **9** (1988), no. 4, 411–422.
14. M. H. Klin, C. Pech, and S. Reichard, *COCO2P, computations in coherent configurations*, Version 0.20, <https://github.com/chpech/COCO2P/>, Sep 2022, GAP package.

15. M. H. Klin, C. Pech, S. Reichard, A. Woldar, and M. Ziv-Av, *Examples of computer experimentation in algebraic combinatorics*, Ars Math. Contemp. **3** (2010), no. 2, 237–258.
16. M. H. Klin and S. Reichard, *Construction of small strongly regular designs*, Tr. Inst. Mat. Mekh. **19** (2013), no. 3, 164–178.
17. A. D. Sankey, *On strongly regular designs admitting fusion to strongly regular decomposition*, J. Combin. Des. **29** (2021), no. 10, 703–722.
18. ———, *Strongly regular designs*, GitHub repository, <https://github.com/sankeyad/strongly-regular-designs>, 2021.
19. A. D. Sankey and C. C. Vanicek, *Some strongly regular fusions of the binary hamming scheme*, in preparation.
20. J. J. Seidel, *A survey of two-graphs*, Colloquio Internazionale sulle Teorie Combinatorie (Roma, 1973), Tomo I, Accad. Naz. Lincei, Rome, 1976, pp. 481–511.
21. N. J. A. Sloane, *A library of hadamard matrices*, <http://neilsloane.com/hadamard/>, Accessed: Aug 2026.
22. D. E. Taylor, *Regular 2-graphs*, Proc. London Math. Soc. (3) **35** (1977), no. 2, 257–274.
23. W. D. Wallis, A. P. Street, and J. S. Wallis, *Combinatorics: Room Squares, Sum-Free Sets, Hadamard Matrices*, Lecture Notes in Mathematics, no. 292, Springer-Verlag, Berlin, Heidelberg, New York, 1972.

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